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Investment Opportunities in the Food Supplements Industry

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Executive Summary

The report assesses the global food supplements industry and identifies the most attractive entry point for a private equity fund.

Because there is no consistent industry taxonomy, we developed a pragmatic three-part classification of the supplements market that is consumer-centric and decision-useful for investors.

- Health & Wellness: largest segment, around €80 billion today, projected to reach approximately €115-120 billion by 2030, implying a CAGR of 6-7%.
- Sports, Performance & Metabolic: mid-sized segment valued around €24.6 billion with stable annual CAGR of 5.92% from 2023-2028 (authors' calculations), driven by mainstream fitness and active lifestyles.
- Beauty & Aesthetic: currently the smallest segment valued around €5.9 billion but growing at a CAGR of 8.97% (2016-2024), supported by beauty and social-media trends.

Market attractiveness is further shaped by regional differences in size, maturity and growth, which create distinct opportunity profiles for the fund.

- Americas: largest and most mature region, with around 6.5% CAGR industry wide.
- Europe: smaller in absolute size but the fastest-growing region, with 9-10% CAGR and a strict regulatory environment with rising per-capita spend.
- Asia-Pacific: structurally growing region, with 6-7% CAGR, supported by middle-class expansion and rapid uptake of beauty and performance products.

On the supply side, there are two dominant business models, with different risk-return profiles.

- Contract manufacturers (CDMOs): segment of around €52 billion, with a 12-13% CAGR and high regulatory and capital barriers.
- Integrated manufacturers: own both production and brands, but are more exposed to brand, channel and execution risk and are less naturally diversified than CDMOs.

To identify potential investments, the report uses an approach of demand side analysis, supply side analysis, and an in depth look of strategies employed by larger players.

- Demand side: Identify the fastest growing and most resilient segment and regions.
- Supply side: Assess the dominant business models with profitable competitive positions by compiling data on main actors by revenue.
- Strategy review: Identify strategies employed by some of the largest firms

- Scenario analysis: Rank investment spaces by structural growth, defensibility, geographic proximity and fit with a European private equity risk-return profile.

The resulting recommendation is to prioritise European Health & Wellness contract manufacturers as the fund's initial entry point into the sector.

- Europe's supplements market is expected to grow from about €24 billion to roughly €44 billion within a decade, with Health & Wellness as the core growth pool.
- Contract manufacturing offers scalable economics with high entry barriers and robust margins, with leading European CDMOs already operating at substantial scale.
- This combination of strong regional growth, attractive CDMO economics, regulatory moats and cultural and geographic proximity makes European Health & Wellness CDMOs the most compelling starting point for the fund.

1. Introduction

The food supplement market has expanded from a specialised niche into a global industry valued in the tens of billions of euros (Statista, 2025a). This growth reflects rising consumer engagement with preventive health, advances in formulation and manufacturing technology and a broad diversification of product formats (Djaoudene et al., 2023). From multivitamins and minerals to plant-based proteins, probiotics and botanical extracts, supplements have become a routine element of contemporary nutrition practices. As uptake continues to increase across regions and demographic groups, the global market offers an informative lens through which to observe wider shifts in health priorities and lifestyle behaviour (Djaoudene et al., 2023).

Food supplements are concentrated sources of nutrients or other substances with nutritional or physiological effects. They are sold as powders, capsules, liquids or tablets and may include vitamins, minerals, amino acids, probiotics or botanical ingredients (FDA, 2023). Their purpose is to complement the diet rather than replace it or serve as a form of disease treatment. Terminology and regulatory frameworks vary across jurisdictions. In the United States, they fall under the Dietary Supplement Health and Education Act of 1994. In the European Union, they are classified as food supplements. In China they are regulated as health foods and in Canada as natural health products (AESGP, 2023)

2. Methodology

The methodology of this report starts from the demand side and is built on a rigorous definition of what we treat as the food supplements industry. Since there is no single global taxonomy and different data providers and regulators use their own category systems, we first reviewed regulatory definitions, market research classifications and real world usage patterns across regions. Based on this review we set clear analytical boundaries for the industry. We include orally ingested non pharmaceutical products that are marketed with a supplementation or functional health benefit, and we exclude conventional foods and pure pharmaceuticals. This core definition, and the way we translate it into a working segmentation of the market, is discussed in more detail in the demand side taxonomy subsection below.

In addition, the report consistently applies the exchange rates reported in the table below, all of which were retrieved from Yahoo! Finance on 29 November 2025 (Yahoo! Finance, 2025).

Table 1. Exchanges Rates per Yahoo Finance		
USD/EUR	**0.862**	**(2025a)**
JPY/EUR	**0.0055**	**(2025b)**
CHF/EUR	**1.072**	**(2025c)**
CNY/EUR	**0.122**	**(2025d)**
INR/EUR	**0.0096**	**(2025e)**

On this foundation we built a coherent analytical framework and then applied it consistently across both the demand side and the supply side. On the supply side we apply a business model focused lens, because revenue alone does not describe the real economics or risk profile of the different types of companies in this industry. We first map the value chain from ingredient suppliers through manufacturing to branded go to market and then group players into a small number of fundamental models. These are ingredient suppliers, contract development and manufacturing organisations, and integrated manufacturers that own both production assets and consumer brands. For each model we combine quantitative indicators such as scale, growth, margin profile, capital intensity and customer diversification with qualitative factors such as the regulatory environment, production and formulation capabilities and the ability to scale across regions and segments. We also examine how these models operate under different regulatory regimes and how they connect to the three demand segments defined below.

By combining this harmonised demand framework with a structured business model analysis of the supply side we can construct a scenario framework in which segment, geography and business model can be compared on a like for like basis. We then use this framework to identify those investment opportunities where attractive structural growth and defensible economics coincide with an acceptable mix of systematic and idiosyncratic risk for a European private equity investor.

2.1. Taxonomy of Supplements

Understanding the supplements market requires an understanding of why different people use different supplements. There are some commonalities between supplements, such as maintaining overall health, strengthening immunity, boosting energy, improving appearance and preventing chronic disease. However, patterns of supplement consumption vary widely across demographic groups, reflecting distinct health priorities and lifestyle objectives. Pregnant and lactating women typically use folic acid, iron and omega 3 fatty acids. Older adults rely on vitamin D, calcium and B complex products to support bone and cognitive health. Children and adolescents commonly take multivitamins to reinforce immunity and growth. Athletes and physically active adults tend to use proteins, amino acids and recovery formulations that enhance endurance and performance (Djaoudene et al., 2023).

Because of this diversity, it is difficult to rely on existing definitions of the dietary supplements market. Commercial datasets and industry reports frequently use inconsistent categorisations. Some combine plant-based formulas with sports nutrition, others separate vitamins from herbal supplements and some merge nutricosmetics into general wellness. These inconsistencies mean that existing definitions are not sufficiently precise or aligned with the functional logic of consumer use (Djaoudene et al., 2023).

To create a more robust analytical foundation, we developed our own taxonomy of the global supplements landscape. Our framework groups products according to the primary consumer need and the underlying mechanism of benefit. This framework corresponds loosely with how some manufacturers categorize supplements (Millmax, n.d.). This creates a coherent and consistent structure for mapping demand, comparing segments and positioning our work within the wider market.

Our first and largest category is the Health and Wellness segment. It includes vitamins, minerals, herbal ingredients and plant-based formulations that target general wellbeing, immune support and nutritional balance. Current market size for this segment is approximately €80 billion in 2024 (Statista, Global health and wellness supplements market size from 2019 to 2030, 2025b).

The second category in our taxonomy is Sports Performance and Metabolic. It covers products that support physical capacity, muscle recovery, metabolism and exercise outcomes. This segment is

currently valued at about €25 billion in 2023 (Statista, Sports and active nutrition market worldwide from 2023 to 2028, 2025c).

The third category is Beauty and Aesthetic, or nutricosmetics. It links supplementation to aesthetic outcomes such as improved skin, hair and overall appearance through ingredients like collagen, hyaluronic acid and antioxidants. The global market size for this segment exceeded €6 billion in 2024 (Statista, Estimated market value of beauty supplements worldwide in 2016 and 2024, 2025d).

Using this taxonomy allows us to describe more accurately how consumer needs structure the global supplements market into three functional clusters and clarifies how each contributes to current total market value. This leaves out some smaller segments, such as pet supplements. Our three categories are focused on capturing the rising culture of preventive self care, increasing interest in personalised and plant-based formulations and the expansion of digital retail channels. Between them, they help explain the continued normalisation of supplement use in everyday health and lifestyle management (Djaoudene et al., 2023; Statista, Total dietary supplements market size worldwide from 2016 to 2028, 2025a).

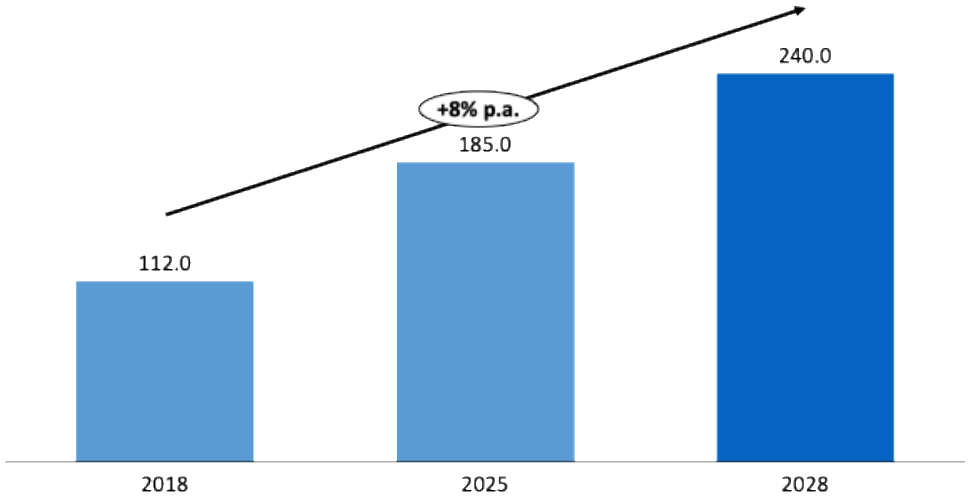
3. Demand

3.1. The Global Food Supplement Market

The global dietary supplements market was valued at approximately €112 billion in 2018, is expected to reach €185 billion in 2025, and projected to rise to €239.4 billion by 2028 (Statista, Total dietary supplements market size worldwide from 2016 to 2028, 2025a). This reflects a compound annual growth rate (CAGR) of around 7.9% from 2018 to 2028. Growth has been driven by consumer awareness of nutrition, wider digital access, and post-pandemic demand for immune-supporting products (Djaoudene et al., 2023).

Differences in geographical distribution stem from socio-demographic and cultural characteristics, consumer behavior, and ageing populations (Djaoudene et al., 2023). North America remains the largest and most mature market. In the United States, the dietary supplement market expanded from €40 billion in 2018 to approximately €58.3 billion in 2024, reflecting a compound annual growth rate (CAGR) of about 6.5%. Furthermore, it is expected to grow up to €66 billion in 2026 (Statista, Total U.S. dietary supplements market size 2016–2024, 2018).

Figure 1. Global Food Supplement Market Growth (in €bn)



Within Europe, the regional market was valued at €23.8 billion in 2019 and is projected to grow at a CAGR of 9.5% to reach €44 billion by 2026. Italy represents one of the strongest national markets, exceeding €1.6 billion in supplement sales and characterized by a pharmacy-led distribution model and high consumer trust (Djaoudene et al., 2023; Farmaimpresa, 2024). Product demand in Italy is concentrated in multivitamins, digestive health products, and herbal formulations.

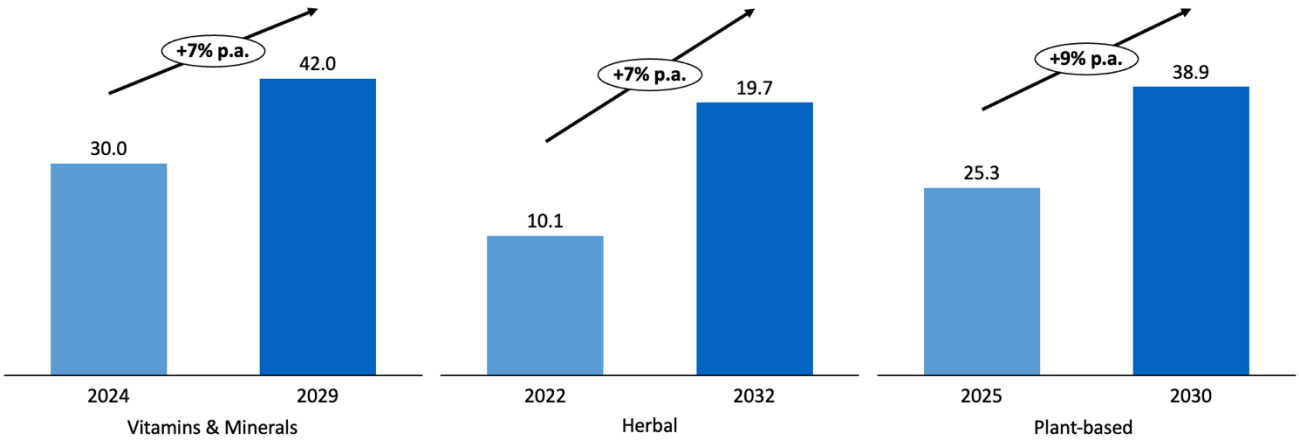
In the Asia-Pacific region, the dietary supplements market has experienced sustained expansion, rising from approximately €29.1 billion in 2018, to €34.2 billion in 2020, and is forecast to reach nearly €48.4 billion by 2026, corresponding to a compound annual growth rate (CAGR) of about 6.5% (Statista, Dietary supplements market value worldwide from 2018 to 2026, by region, 2024a) Growth is supported by increasing disposable income, an expanding middle class, and greater awareness of preventive health. Within the region, Japan and China continue to lead demand, combining long-standing traditions of functional nutrition with rapid innovation in wellness and digital health channels.

3.2. Health and Wellness Supplements

Health and wellness supplements constitute the largest revenue segment within the global dietary supplements market and are the most widely consumed worldwide. According to Statista (2025b) and Djaoudene et al. (2023), around two-thirds of supplement users take a vitamin or mineral product, underscoring the central role of this category in global supplementation habits. These products, spanning vitamins, minerals, herbal, and plant-based ingredients, support overall health, immune function, and nutrient adequacy, positioning them as a backbone of preventive health routines.

Within this landscape, the worldwide Vitamins & Minerals segment reached approximately €29.7 billion in 2024 and is projected to rise to about €42 billion by 2029, a ~7% CAGR (Statista, 2024b). In parallel, herbal and plant-based supplements are expanding rapidly as consumers seek natural and sustainable options: the global herbal dietary supplements market was about €10.1 billion in 2022 and is projected to reach €19.7 billion by 2032 (~7.1% CAGR) (Allied Market Research, 2021), while the plant-based supplements market is forecast to grow from €25.3 billion in 2025 to around €38.9 billion by 2030 (~9% CAGR) (Markets and Markets, 2025). Together, these segments show a diversified health and wellness supplements industry, balancing a strong vitamin/mineral core with fast-growing herbal and plant-based categories.

Figure 2. Health & Wellness Supplement Market Growth, By Segment (in €bn)



Demographic shifts are also reshaping demand patterns. McKinsey's *Future of Wellness* (2025) report highlights that Millennials and Generation Z consumers, while representing just over one-third of the global adult population, account for more than 40% of total wellness spending. These younger cohorts are driving market growth through their willingness to experiment with new formats and digitally enabled experiences. They show a stronger preference for personalized, science-backed, and ethically sourced products, often discovered via social media or influencer networks rather than traditional retail channels. This generational transition is accelerating innovation cycles and redefining how supplement brands communicate credibility, efficacy, and trust.

Beyond regional and generational differences, innovation in format and functionality is reshaping consumer engagement within this subcategory. In the U.S., for example, non-pill supplement forms, such as gummies, powders, and ready-to-drink liquids, now account for nearly two-thirds of market share (Nutraceuticals World, 2024). At the same time, the thematic focus of new product launches has

expanded from general wellness toward mental health, mood, and cognitive performance, with brain and mood health claims growing at annual rates of 12 to 22% (Innova Market Insights, 2024). The global "mental well-being supplements" segment alone is valued at approximately €11 billion in 2025 and is projected to grow at an 6.3% CAGR to reach €14.9 billion by 2030 (Research & Markets, 2025). This diversification of product forms and claims reflects a convergence between nutrition, lifestyle, and mental well-being, broadening the scope of the health and wellness segment beyond traditional vitamin and mineral formulations.

Consumer motivation research across Europe adds further nuance to this shift. According to the Food Supplement Europe (FSE) Consumer Survey 2022, conducted across 14 EU countries, the majority of supplement users cite maintaining overall health, supporting immunity, and compensating for nutrient deficiencies as their main reasons for purchase. Trust in pharmacists and doctors as key information sources remains high, especially in Southern Europe, while younger consumers increasingly rely on social media and digital health platforms for product discovery. This dual reliance on medical authority and digital community underscores how the supplements market is blending traditional trust channels with new, tech-enabled modes of engagement.

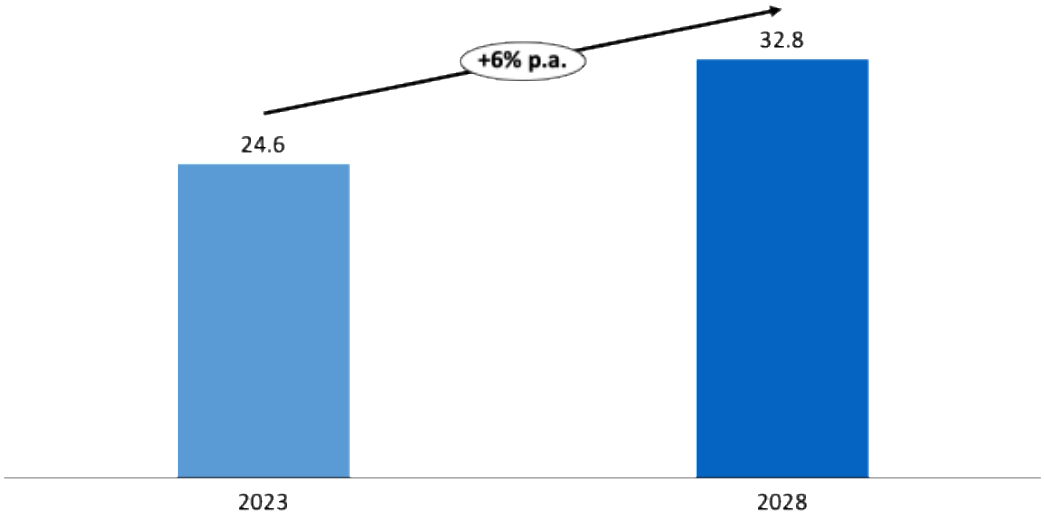
Across all regions, the health and wellness supplements segment demonstrate notable resilience, supported by structural drivers, including global population aging, persistent micronutrient deficiencies, and post-pandemic immunity focus (Djaoudene et al., 2023). However, emerging consumer trends are also transforming its future trajectory, including the rise of personalized nutrition, where DNA-based and AI-driven recommendations guide supplement choices, the integration of digital health ecosystems, linking wearable data with supplement tracking, and an accelerating focus on longevity and cognitive vitality, particularly among aging but health-conscious populations (McKinsey, *Future of Wellness Trends*, 2025).

Together, these forces consolidate health and wellness supplements as both the economic and behavioral core of the global dietary supplements market, the largest and most widely consumed segment worldwide, now evolving rapidly through innovation, digitalization, and a broader vision of holistic well-being.

3.3. Sports, Performance & Metabolic Supplements

Sports, performance and metabolic supplements serve three core consumer motives: building strength and lean mass, supporting energy and recovery for training, and managing body composition and weight. These needs benefit from the wider growth of dietary supplements. Within this, the sports and active nutrition segment is expected to grow from €24.6 billion in 2023 to €32.8 billion by 2028, which translates into a CAGR of 5.92%. This signals persistent demand beyond the pandemic period (Statista, 2025c). Formats and usage patterns also support the mainstream position of the category.

Figure 3. Sports and Active Nutrition Supplement Market Growth (in €bn)



In the United States powders and formulas have been the dominant format, holding 84.1% share in 2015, compared with 11.4% for ready to drink beverages and 4.5% for pills. This reflects the central role of shakes and mixable products in gym culture and in planned replenishment cycles, where larger pack sizes and stackable regimens drive value (Statista, 2015). Diet behaviour further reinforces performance and metabolic usage, since in 2025 the most followed approach in the United States was high protein at 23%, ahead of mindful eating and intermittent fasting, a pattern that aligns naturally with protein powders, creatine, and hydration or amino products and that supports weight control solutions around training plans (Statista, 2025e).

Across Europe, including Italy as a clear case, consumers show a protein first spending pattern inside sports nutrition. A cross country view of per capita spending in 2019 indicates that protein shakes and high protein snacks are the largest line items in many markets, followed by vitamins and minerals and then weight loss. In Italy, annual per capita spend reached €378 on protein shakes and €245 on high

protein foods and snacks, alongside €225 on vitamins and minerals and €94 on weight loss products, which suggests that performance goals are pursued inside wider wellness routines rather than as stand alone hacks (Statista, 2025f).

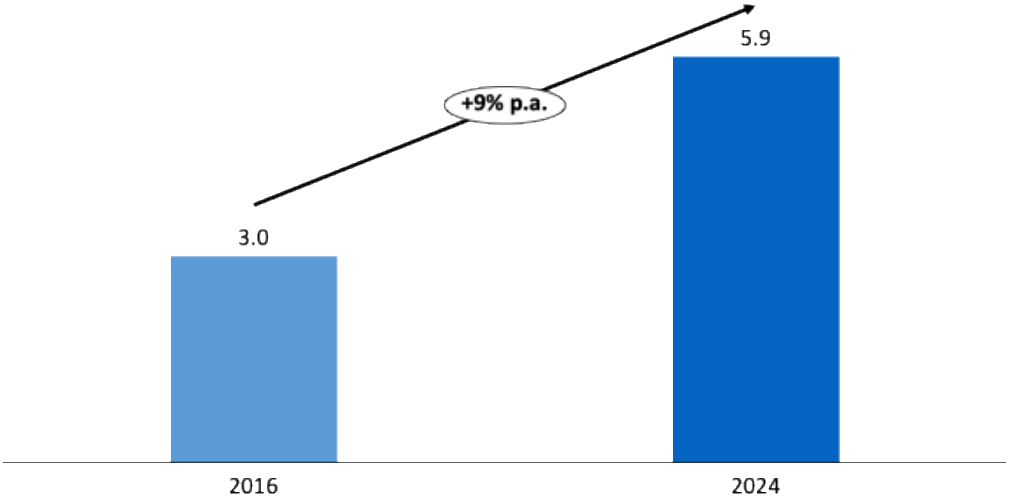
In Asia, weight management remains a persistent theme that boosts cross sell for metabolism products. In China, high vitamin use with proteins and minerals also ranking high creates a large addressable base for performance products, while digital commerce and social selling expand trial for protein, amino and energy formulas (Statista, 2025g). Japan's retail and regulatory infrastructure makes daily nutrition mainstream for healthy aging and performance. Convenience stores and vending networks promote ready to drink protein, amino acid and electrolyte drinks, alongside capsules and tablets (Djaoudene et al., 2023).

Across markets, the category's growth rests on three reinforcing facts that now look structural. First, protein use has moved into the mass market and ranks just behind vitamins, which signals everyday adoption of performance goals rather than niche behaviour (Statista, 2025h). Because powders dominate key markets, their format allows for planned replenishment, larger package sizes, and bundling with complementary ingredients such as creatine and electrolytes, all of which drive repeat purchases and increase customer lifetime value (Statista, 2015). Third, weight goals remain a strong and stable need, confirming that performance and metabolism objectives are often pursued together in the same purchase.

3.4. Beauty & Aesthetic Supplements

Beauty focused ingestible supplements such as collagen, hyaluronic acid, biotin and hair skin nail formulas sit at the junction of beauty care and wellness. The category benefits from the broader expansion of dietary supplements, which provides scale, shelf access and marketing support for beauty use cases as well (Statista, *Total dietary supplements market size worldwide from 2016 to 2028*, 2025a). Within this tailwind, the beauty supplements market nearly doubled from about €3.0 billion in 2016 to about €5.9 billion in 2024, confirming sustained consumer interest in aesthetic outcomes delivered through ingestibles (Statista, 2025d).

Figure 4. Beauty Supplement Market Growth (in €bn)



Consumer demand is shaped by three reinforcing drivers. First, the convergence of beauty and wellness lifts the appeal of beauty from within, where consumers seek visible skin and hair benefits through daily routines rather than episodic treatments (Euromonitor International, 2024). Second, the category's formats and usage patterns fit subscription like behaviour. Collagen and skin hydration products are typically taken over multi month cycles, which favours larger pack sizes, replenishment and regimen bundling with complementary actives. Third, discovery and conversion lean on digital channels and social proof while pharmacy and drugstore shelves scale mainstream access, creating a two-speed go to market well suited to education intensive products (Euromonitor, 2024).

Signals from Asia illustrate how beauty goals have become mainstream within supplements. In Southeast Asia in 2024, beauty supplements accounted for 26.8% of the dietary supplements category, indicating that skin and hair objectives are now a common part of the basket rather than a niche add on (Statista, 2025i). Read together with global market value growth for beauty ingestibles, these regional patterns support the view that the category is crossing from trend to habit (Statista, 2025d).

Overall, beauty and aesthetic supplements are a relatively small but fast growing slice of the dietary supplement universe. The combination of a rising global base, clear willingness to commit to routines that promise visible results, and retail plus digital reach suggests a resilient outlook. With the market value nearly doubling between 2016 and 2024 and with share indications that beauty goals already command meaningful space within supplement baskets (McKinsey, 2025b).

4. Supply

This section discusses the supply chain in full. As will be seen, while supply chains in the supplements industry have special characteristics in comparison to other markets, our different categories are very similar in structure in comparison to each other. Most of the heterogeneity lies in regulations across regions. As such, this section begins with outlining regulatory environments, followed by a description of the supply chains, outlining some specific traits.

Two distinct types of firms emerge. Firstly, vertically integrated firms that encompass sales to consumer and retail, manufacturing, and sometimes production of raw ingredients. Secondly, firms specializing in individual steps of the ingredients or manufacturing processes. From the second category, contract manufacturers emerge as key industry actors. Tables present the actors in each category with the highest revenues. Finally, we present the case for why investors should consider contract manufacturing the dominant industry segment and pay it extra attention in investment decisions.

4.1. Regulatory Environments

4.1.1. US Regulations

In the US, supplements are primarily regulated by two main federal agencies – the Food and Drug Administration (FDA) for supplement content and manufacturing practices, and the Federal Trade Commission (FTC) to ensure fair marketing practices (CRN, n.d.). In supplements the responsibilities of the agencies overlap somewhat but are clearly regulated in a 2017 memorandum of understanding between the two (FTC 2022; see FDA, 2017).

The FDA regulates both food and medicines, which conveniently sidesteps issues with whether supplements are to be considered as food or medical drugs. However, the FDA does consider most supplements to be under the umbrella of foods (FDA, 2024). Sellers generally do not require direct approval from the FDA to sell their product, unless a new, non-reviewed compound is being used in it (FDA, 2024). Sellers are subject to abide by regulations stipulated in the FD&C act. Assuming no ingredients used have not received FDA approval, its regulations approximate fair trade practices. The seller is responsible for ensuring its products are not misbranded or adulterated before final sell, or that they are not otherwise in violation of federal law. (FDA, 2024).

On advertising, authority is split between the FDA and FTC (FTC 2022; see FDA, 2017). In brief, the FDA ensures proper branding, while the FTC regulates claims made about the efficacy of supplements

(FDA, 2017). For instance, sellers must disclose ingredients use in their product, and if a seller claims a specific supplement benefit, it must have evidence backing their claim (FDA, 2024).

US regulations are seen by some experts as comparatively weak. The congressional research services summarises that "FDA generally does not take regulatory action on food or dietary supplements until something goes wrong with a product that is on the market" (CRS, 2021: p2). On the other hand, the current US secretary of health and human services, Robert F. Kennedy Jr, describes targets the "aggressive suppression of dietary supplements", indicating the federal government might seek to loosen regulations in the future (Allen, 2025).

4.1.2. EU Regulations

Unlike the comparatively straightforward American regulations, EU regulations are significantly more complicated. A comprehensive review of sports supplements (Varvaštian, 2015) places them under the food supplements umbrella, which is in turn subject to both broad and narrow laws. Beauty supplements are even more ambiguous, as the concept is poorly defined. Beauty supplements rarely occur as a seperate category in legal documents, being considered more alike food supplements with beauty benefits (Food Supplements Europe, 2014). A full review of these articles is beyond the scope of this report.

It can be summarized that EU laws are very strict in comparison to American ones. For instance, for any food supplement to make a claim about its impact on a user's health, it must get clearance from the European Food Safety Authority (EFSA, 2025). Compare this with the aforementioned US approach of only taking action once "something goes wrong" (CRS, 2021: p2), in contrast to which EU regulations appear much stricter.

The regulatory landscape in the EU is a potentially major barrier to entry for new entrepreneurs. Possibly as a response to the EU's strictness and opaqueness on health claim regulations, supporting industries have emerged. There are for instance legal services that aid supplement entrepreneurs with the regulatory process, in filing health claims (FoodChain ID, n.d.) It is also noteworthy that contract manufacturers in our database, who often alleviate significant regulatory burden on part of entrepreneurs (Eubioco, 2025) are EU-heavy in our database, with five of the eight contract manufacturers being based in regulation heavy Europe (table 2).

In 2024, members of the European Parliament called for reforms to the health claims system (EUR-Lex, 2024), indicating that reforms the claims system may occur. Further developments are yet unclear, but might result in a loosening of regulations, creating new opportunities on the EU market.

4.1.3. China Regulations

In China, dietary supplements are regulated under the broader category of "health food" (保健食品), which includes products that claim specific health functions or are intended to supplement vitamins and minerals (NuFFooDS Spectrum, 2023). Oversight is managed primarily by the State Administration for Market Regulation (SAMR) (CIRS Group, 2025). The regulatory framework distinguishes between two approval pathways: registration and filing (notification). Products that use approved ingredients from the national Health Food Raw Material Directory and make only nutrient-related claims, such as vitamins and minerals, follow the simplified filing procedure, while those that include new ingredients, non-listed functional claims, or are imported typically require full registration (Freyr Solutions, 2024). The registration process is extensive and can take 12 to 24 months, with certificates usually valid for five years before renewal (CIRS Group, 2025).

All ingredients must comply with the approved raw material lists, and products containing novel ingredients require additional evaluation or new raw material registration (Ettinger & Li, 2025). Imported supplements must also meet Chinese national food safety standards (GB standards) and can only be distributed after clearance by the General Administration of Customs of the People's Republic of China (GACC) (Shanghai Zhongshen, 2024). Labels must be in Chinese and include the product name, approved functions, ingredient list, country of origin, manufacturer and importer details, and storage instructions (NuFFooDS Spectrum, 2023). Only officially approved health claims, currently 27 recognized types, are permitted, and marketing materials must avoid any suggestion of disease prevention or treatment in accordance with the Advertising Law of the People's Republic of China (Freyr Solutions, 2024).

Manufacturers must operate under recognized GMP or food safety management systems and demonstrate compliance during audits and site inspections (Ettinger & Li, 2025). Foreign firms must appoint a local Chinese agent or legal representative for the registration process, as only one certificate owner per product is allowed (CIRS Group, 2025). For imported supplements, proof of prior commercialization in the country of origin is often required (Shanghai Zhongshen, 2024). The system ensures product traceability and accountability across the supply chain (NuFFooDS Spectrum, 2023).

4.1.4. India Regulations

In India, dietary supplements and nutraceuticals are regulated by the Food Safety and Standards Authority of India (FSSAI) under the framework of the Food Safety and Standards Act, 2006 and the Food Safety and Standards (Health Supplements, Nutraceuticals, Food for Special Dietary Use, Food for Special Medical Purpose, Functional Food and Novel Food) Regulations, 2016 (Food Safety and Standards Authority of India, n.d.). Products must be manufactured, packaged, distributed, and sold only after obtaining the appropriate FSSAI license or registration (Pharma Regulatory, 2025). Ingredients used must be from approved schedules defined under the regulations, and product formats such as capsules, tablets, or powders must comply with recognized monographs such as those in the Indian Pharmacopoeia (Certification-India, 2024). Labelling and packaging are governed by the Food Safety and Standards (Labelling and Display) Regulations, 2020 and the Food Safety and Standards (Packaging) Regulations, 2018, which outline mandatory declarations and language requirements (Food Research Lab, 2025). Labels must be in English or Hindi and must declare the list of ingredients, nutritional information, dosage instructions, batch number, expiry date, manufacturer name, and FSSAI license number (Food Research Lab, 2025).

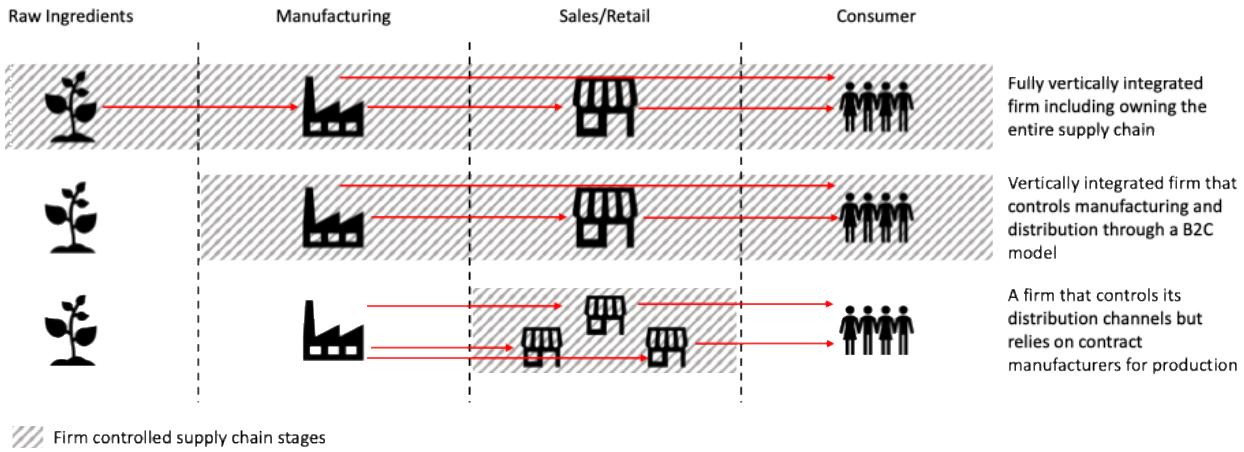
Claims made for health supplements must follow the list of permitted claims and cannot present the product as a treatment or cure for disease (Food Safety and Standards Authority of India, n.d.). Imports of dietary supplements are subject to FSSAI import rules, and overseas manufacturers must meet GMP standards (Pharma Regulatory, 2025). A local Indian agent or representative is often required for import and licensing processes (Pharma Regulatory, 2025). All imported products must comply with Indian food safety and additive standards before entering the market (Certification-India, 2024). The regulatory environment in India is evolving. Draft amendments introduced in 2022 proposed stricter oversight of novel ingredients, health claims, and manufacturing standards to align the supplement sector more closely with pharmaceutical-grade practices (Certification-India, 2024). This reflects India's growing emphasis on safety, traceability, and scientific validation within the nutraceutical and supplement supply chain (Food Research Lab, 2025).

4.1.5. International Standards

Internationally, supplement manufacturers are expected to follow globally recognised quality and safety standards that ensure consistency, traceability, and regulatory compliance across markets. The most widely accepted benchmark is Good Manufacturing Practice (GMP), which establishes strict requirements for hygiene, quality control, and documentation throughout all stages of production, from raw material sourcing to packaging. Many countries align their GMP systems with World Health Organization (WHO) guidelines, making it the universal foundation for supplement manufacturing and export readiness. Beyond GMP, ISO certifications provide further assurance of quality and safety, with ISO 9001 focused on quality management systems and ISO 22000 addressing food safety controls, including hazard analysis and critical control points (HACCP). The FSSC 22000 certification builds on ISO 22000 and is recognized by the Global Food Safety Initiative (GFSI), offering the highest level of food safety assurance and often required by global retailers and distributors. International associations such as the International Alliance of Dietary/Food Supplement Associations (IADSA) and Food Supplements Europe also promote harmonised Good Manufacturing Practice frameworks specific to supplements, covering formulation, production, packaging, storage, and distribution. Collectively, these global standards ensure product quality and safety, enable easier cross-border compliance, and strengthen consumer trust in supplement brands operating across international markets.

4.2. Supply Chain Overview

Figure 5. Supply-Chain Overview



4.2.1. R&D and Raw Materials Sourcing

R&D consists of, among other things, identifying specific consumer needs (Pet Tech Labs, n.d), recipe development and fine tuning of specific products (Fowler, 2022), readjustment to an existing formula (RSSL, n.d.), among others.

A notable feature is the close interplay between R&D and raw materials sourcing, as the quality of raw materials are an exceptionally integral part of product quality. There are instances where disruptions of specific ingredients in a product had to be solved by a reworking of the underlying formula (see RSSL, n.d). As such, R&D must work closely with materials sourcing.

The sensitivities of raw material sourcing vary somewhat across product type. Sports supplement consumers are sensitive to substitution of ingredients (Fowler, 2022). An article describing the manufacturing process for nutraceuticals highlights the importance of organic rather than synthetic ingredients (Ullrich, 2025), while an article on the wider beauty industry emphasizes ethical sourcing of materials (CentricSoftware, 2025).

4.2.2. Contract and Integrated Manufacturing

Contract manufacturing is very prominent, if not dominant, in the supplements industry. Firms like Millmax (n.d.) and Eubico (2025) provide a contract manufacturing service for prospective entrepreneurs. The often prohibitively high costs of establishing manufacturing operations, often associated with regulatory compliance and capital investment, are sidestepped by smaller entrepreneurs through use of contract manufacturers (Eubico, 2025).

Because contract manufacturers are not involved in B2C sales, their importance is easily overlooked in pure company valuations in comparison to vertically integrated firms. For instance, a major contract manufacturer like Aenova has annual sales at just under €0.86 billion (see table 2), in comparison to Herbalife at €4.39 billion (Herbalife, 2024). Nonetheless, Aenova is Europe's largest provider of solid and semi-solid supplements, and the world's second largest semi-solid supplements supplier (Aenova, 2025).

Taken together, contract manufacturing is a major and fast-growing subsector. Grand View Research puts global market size of supplements in the wider sense at €165.68 billion with a CAGR of 8.9% through 2023 (2025a). Contract manufacturing alone stands for €51.28 billion, with a far higher CAGR of 12.6% annual through 2030 (2025b).

On the other side, larger and smaller firms maintain in house manufacturing. These can be small boutique shops, such as the MLM firm Nature's Sunshine (2024), but larger firms with integrated production do exist. Some large actors are American Amway, with sales at €6.62 billion across all categories (Amway, 2024), American Herbalife with sales at €4.39 billion, and Chinese By-Health with sales of €816 million (Wall Street Journal, 2025).

4.2.3. Distribution and Sales

Contract and integrated manufacturing is involved in two radically different models of distribution and sales. Integrated manufacturers sell to retail, and consumers via websites. The integration is itself a selling point, often branded to consumers through ensured quality and traceability (see Amway, n.d. and Nature's Sunshine, 2024). As mentioned, contract manufacturers supply practically no product themselves, instead selling their product to other entrepreneurs for branding and distribution. These can be anything from established actors to influencers with their own brands of supplements. Aenova alone serves more than 400 different entrepreneurs (Aenova, 2025). An understanding of the strategies of distribution entrepreneurs is elaborated upon in the business strategy section of this report.

It is also important to note that two of the largest retail-focused supplement companies, such as Herbalife and Amway, operate through multi-level marketing (MLM) compensation structures ((Herbalife Ltd., 2024) and (Amway 2024)). While these firms are among the biggest revenue generators in the global supplements space, a substantial number of smaller MLM organisations also participate in the industry, even if they do not appear among the leading players. This distribution model is therefore a recurring feature of the supplements market and is worth bearing in mind when considering the breadth and nature of commercial activity within the sector.

4.3. Table of General Big players + Retailers

Table 2. Overview of General Biggest Market Players

Company	Segment (Ingredient / CDMO / Capsules & Delivery Tech)	Estimated Revenue (FY 2023–24)	Category Fit (H&W; Sports/Metabolic; Beauty/Aesthetic)	Core Formats/ Tech	Geography - HQ	Geography - Manufacturing Footprint
Aenova	CDMO–contract manufacturer	≈ €833 million	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Tablets, Capsules, Powders, Liquids	Germany	Germany, Switzerland, Italy, Ireland, Romania, USA
Sirio Europe (Sirio Pharma)	CDMO–contract manufacturer	≈ €436.9 million	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Softgels, Gummies, Tablets, Probiotics	Germany	China, USA, Germany
Lonza Capsules & Health Ingredients	Capsules & delivery tech	€1.032 billion	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Capsules, HPMC Capsules, Probiotics	Switzerland	Switzerland, USA, China, Japan, India
DSM-Firmenich	Ingredient supplier	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Premixes, Vitamins, Nutraceuticals	Switzerland / Netherlands	Global (EU, US, APAC)
Rousselot (Peptan)	Ingredient supplier (collagen)	≈ €1.46 billion (2023, Darling Ingredients Food Ingredients segment)	Health & Wellness, Beauty/Aesthetic	Collagen Peptides	France	France, Belgium, Brazil, China, USA
Millmax	CDMO–contract manufacturer	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Capsules, Tablets, Powders, Gummies	Poland	Poland
Eubioco	CDMO–contract manufacturer	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Tablets, Capsules, Powders, Sachets	Poland	Poland
BASF Nutrition & Health	Ingredient supplier	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Vitamins, Carotenoids, Plant Sterols	Germany	Germany, Global network
Vitaquest	CDMO–contract manufacturer	≈ €215–258 million (estimate)	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Capsules, Tablets, Powders, Gummies	USA	USA (NJ facilities)
NutraScience Labs	CDMO–contract manufacturer	≈ €86–129 million (estimate)	Health & Wellness, Sports/Metabolic	Capsules, Powders, Gummies, Tablets	USA	USA
Kaneka Nutrients	Ingredient supplier	Non-public	Health & Wellness, Sports/Metabolic	CoQ10, Ubiquinol	Japan / USA	USA (Texas), Japan
Glanbia Nutritionals	Ingredient supplier	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Proteins, Bioactives, Premixes	Ireland / USA	Ireland, USA, China
TSI Group	CDMO–contract manufacturer	≈ €172–215 million (2023 estimate)	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Capsules, Powders, Tablets, Custom Formulations	China / Australia	China, Australia, Thailand, EU, USA
Novonesis (Novozymes + Chr. Hansen)	Ingredient supplier	Non-public	Health & Wellness, Beauty/Aesthetic	Probiotics, HMOs, Enzymes	Denmark	Denmark, USA, Global
Kerry ProActive Health	Ingredient supplier	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Probiotics, Postbiotics, Bioactives	Ireland	Global
Nitta Gelatin / Wellnex	Ingredient supplier (collagen & gelatin)	€308 million	Health & Wellness, Beauty/Aesthetic	Collagen Peptides, Gelatin	Japan	Japan, USA, India

Note: sources are found in Appendix 1.

The table provides a consolidated overview of the major supply-side players operating within the global food supplements industry, organised by company segment, revenue, category focus, technological formats, geographical footprint, and certification status. It includes both large multinational ingredient suppliers and specialised contract development and manufacturing organisations (CDMOs), highlighting the variety of business models that support the supplement value chain. Each company is assessed according to its alignment with the three core supplement categories, namely Health and Wellness, Sports and Metabolic, and Beauty and Aesthetic, together with its technical competencies, which range from capsule and softgel manufacturing to probiotics, collagen peptides, and advanced nutritional premixes. The table also distinguishes between headquarters location and international manufacturing sites, illustrating the wide geographical distribution of production capacity across Europe, North America, and Asia. Revenue data, where publicly available, provides an indication of relative market scale, with clear differences between global leaders such as Rousselot, Lonza, and Aenova, and the smaller regional CDMOs whose financial information is not disclosed. Finally, the inclusion of certifications such as GMP, ISO 9001, ISO 22000, and FSSC 22000 underscores the level of regulatory compliance and operational maturity across the listed organisations, offering a comprehensive snapshot of the structural landscape shaping supplement manufacturing.

Table 3. Overview of Biggest Market Players – Retail Focus

Company	Supplement Revenue (Latest)	Manufacturing activity	Manufacturing Locations	Retail Focus	Additional Information	Category Fit	Headquarters
Nestlé Health Science	€5.59B (2024)	Yes	USA, Canada, France, Germany, Brazil, China, Switzerland	Global retail brands, pharmacies, e-commerce, grocery	Owns Garden of Life, Vital Proteins, Pure Encapsulations	Health & Wellness, Beauty/Aesthetic, Sports/Metabolic	Vevey, Switzerland
Herbalife Nutrition	€4.27B (2024)	Yes	USA, Mexico, Brazil, China, India	Direct-to-consumer retail, global distributors	Global sports & active nutrition brand	Health & Wellness, Sports/Metabolic	Los Angeles, USA
Amway (Nutrilite)	€2.92–3.18B (2024)	Yes	USA (California, Washington), Brazil, Mexico, India, China	D2C/MLM retail distribution	Vertically integrated; operates own farms	Health & Wellness, Sports/Metabolic	Ada, Michigan, USA
Haleon (Centrum, Caltrate)	€2.67B (2024, VMS only)	Yes	USA, UK, Ireland, China, India	Pharmacies, mass retail, supermarkets	Large global consumer-health portfolio	Health & Wellness	Weybridge, United Kingdom
Bayer VMS	€1.46B (2024, VMS only)	Partially	Germany, USA, Spain, South America	Pharmacies, grocery, mass retail	Brands include One A Day, Berocca, Elevit	Health & Wellness	Leverkusen, Germany
Pharmavite (Nature Made)	€1.29B (2024)	Yes / Partly	USA (California, Ohio)	Pharmacies, mass retail, e-commerce	Top US vitamin brand owned by Otsuka	Health & Wellness	West Hills, California, USA
Church & Dwight (Vitafusion)	€946M (2024, VMS)	Yes	USA	Mass retail, pharmacies, grocery	Leader in gummy supplements	Health & Wellness	Ewing, New Jersey, USA
Abbott Nutrition (Supplements)	€1.20B (2024 est.)	Yes	USA, Ireland, Netherlands, Spain, Singapore	Pharmacies, grocery, clinical retail	Includes Ensure and Pedialyte	Health & Wellness	Abbott Park, Illinois, USA
NOW Foods	€645M (2024)	Yes	USA (Illinois, Nevada)	Natural retailers, mass retail, e-commerce	Large clean-label supplement manufacturer	Health & Wellness, Sports/Metabolic	Bloomingdale, Illinois, USA
Swisse (H&H Group)	€499M (2024)	No	Manufactured via CDMOs in China, Australia, EU	Premium retail, pharmacies, e-commerce	Beauty and wellness positioning	Beauty/Aesthetic, Health & Wellness	Hong Kong / Melbourne, Australia

Note: sources are found in Appendix 1.

4.4. Why Contract Manufacturing Should Be the Investor's Main Focus

When analysing large players, other reports have highlighted different main actors than we do. For instance, Grand View Research (2025a) mention company like Pfizer Inc., GlaxoSmithKline, and others. These are assumedly chosen based on pure market size. However, through our supply-demand framework we have come to a different conclusion. Contract manufacturing, while individually smaller by financials, have an indirect market impact potentially exceeding the traditional giants. This section argues why contract manufacturers should be the investor's main focus.

For large pharmaceutical companies such as Pfizer, supplements are a small part of their operations. Their main strategic focuses as regulated medical manufacturers do not align with the positioning, regulatory environment, or commercial priorities of the mainstream dietary supplements industry. As

highlighted in reporting from NBC News (Brandon, 2009), major pharmaceutical companies often choose not to participate directly in the supplements sector because doing so contradicts their identity as scientifically rigorous and clinically validated healthcare producers. The supplements market operates with considerably lower regulatory requirements. This regulatory disparity creates a reputational risk for pharmaceutical firms, as association with lightly regulated supplements could undermine the trust they have built through evidence-based drug development and professional medical oversight. As such, they either distance themselves from these markets, or act via smaller company subdivisions.

Table 3 presents a set of retail-focused supplement giants, including company subdivisions like Nestlé. These firms have commercial models, brand portfolios, and product strategies rooted in the consumer supplements space; these companies often generate significantly higher revenues than manufacturing- or ingredient supply-firms and therefore offer meaningful insight into the global supplements landscape.

In comparing tables 2 and 3, contract manufacturers and ingredient suppliers are smaller in revenue. This is a natural consequence of these firms not being directly in sales to consumers, missing out on the value added. However, this hides the market power of these "hidden giants". Aenova alone serves more than 400 clients and are the largest solid and semi solid supplements manufacturer in Europe (Aenova, 2025). The decisions of an integrated firm like Nutralite would have market implications, but Aenova's decision has a direct impact on many hundreds of firms both up- and down-stream.

Viewed as a whole, contract manufacturing becomes the dominant force in the industry. Adding up revenues of all the major actors in table 3 amounts to just over €21.5 billion. These firms sell to consumers, meaning they benefit from consumer markups. Contract manufacturing, by contrast, stands for €51.28 billion in 2025, with a CAGR of 12.6% through 2030 (2025b) only for the manufacturing part of the industry. If the value adding customer markups could be considered, it is almost plausible that their impacts could approach a majority of the €165.68 billion market (2025a). As such, between their immense impact on the industry and their high CAGR, contract manufacturers are the most dynamic, most important, and fastest growing segment of the supplements market. The strategy part of this document as such analyses their strategic moves, alongside the ingredient producers used by them, and the integrated firms.

5. Analysis of Largest Players

5.1. Rousselot (Peptan, Darling ingredients)

Rousselot the first biggest player is an Ingredient Supplier operating in the categories of Health & Wellness and Beauty & Aesthetic industries. They are focused on the production of Collagen Peptides for joint, bone, and beauty-from-within applications under the globally recognized Peptan brand. Their Headquarters are in France, and they maintain a large international manufacturing network in France, Belgium, Brazil, China, and the USA. Rousselot operates as part of Darling Ingredients and represents one of the company's most strategically significant divisions. The firm leads the global collagen market through continuous innovation, scientific validation, and sustainable sourcing initiatives. It is widely regarded as a benchmark for quality and traceability within the collagen sector, supplying both major consumer brands and specialized manufacturers. They reach a yearly revenue of approximately €1.46 billion in the 2023 Food Ingredients segment.

The company operates as a global B2B ingredient supplier, creating value through the large-scale production and commercialization of collagen and gelatine for downstream manufacturers. Operating under the Peptan brand, Rousselot state that they differentiate themselves through scientific validation, traceability, and sustainability being core components of their value proposition (Darling Ingredients, 2025). Its model emphasizes industrial scale and R&D excellence, supported by 16 production facilities across four continents that enable reliable global supply (Darling Ingredients, 2025).

Rousselot's revenue streams are primarily generated through its position as ingredient supplier within Darling Ingredients' Food Ingredients segment, which reported €1.46 billion in net sales in 2023, accounting for approximately 25% of the group's total revenues (Darling Ingredients, 2024). Its operating income of ≈ €182 million reflects a robust segment margin of around 12.5%. The firm's ongoing investment in R&D, sustainability certifications, and tailored ingredient solutions further supports premium pricing and customer retention across diverse geographic markets (Darling Ingredients Form 10-K, 2023).

As an upstream ingredient manufacturer, Rousselot's relationship with the retail platform Amazon is indirect. The company does not sell its products directly to end consumers via Amazon, instead many of its B2B clients, market their supplements on Amazon. So, while Amazon plays an important role in the global distribution of collagen-based supplements, it's not part of Rousselot's own distribution or revenue strategy.

Customer acquisition and retention in Rousselot's model depend on brand reputation, scientific credibility, and long-term partnerships. Peptan functions as a high-quality brand and recognized by manufacturers worldwide (Darling Ingredients, 2025). While continuous R&D investment strengthens customer loyalty by supporting innovation in new applications such as sports recovery and gut health (Nutrition Insight, 2023). By combining scientific rigor, technical collaboration, and consistent quality, Rousselot has established itself as a preferred supplier in both the Health & Wellness and Beauty & Aesthetics supplement categories.

5.2. Lonza Capsules & Health Ingredients

Lonza Capsules & Health Ingredients operates as a Capsules and Delivery Technology Supplier serving the Health & Wellness, Sports & Metabolic, and Beauty & Aesthetic categories. The division is focused on the production of capsules, HPMC capsules, and probiotic formulations, most notably through its Capsugel platform. Headquartered in Switzerland, it maintains a global manufacturing footprint across Switzerland, the USA, China, Japan, and India. Lonza serves both pharmaceutical and nutraceutical industries, offering precision-engineered capsule solutions designed for stability, bioavailability, and targeted nutrient release. The division reported annual sales of approximately €1.13 billion in 2024(Lonza, 2025). In December 2024, Lonza's Board of Directors officially announced its plan to exit the CHI business, a strategic move to focus on its core pharmaceutical CDMO business. This classifies the division as a non-core asset, implying a potential risk of de-prioritization in R&D investment and capital allocation, which could slow its response to emerging market trends.

Revenue for the Lonza CHI division is generated by supplying over 7,000 B2B customers across the pharmaceutical and nutraceutical sectors. For the 2024 fiscal year, the division generated sales of €1.13 billion and a CORE EBITDA of €1.13 billion, representing a robust 24.3% margin (Lonza, 2025). This profitability was defended through operational efficiencies despite market destocking pressures Revenue streams are derived from three core businesses: Hard Empty Capsules, led by the flagship Capsugel brand; Innovative Dosage Form Solutions, an end-to-end contract manufacturing (CDMO) service for nutraceutical clients; and Active Lifestyle Health Ingredients, which sells science-backed branded ingredients such as UC-II for joint health and Carnipure L-carnitine.

Customer attraction for the Lonza model is based on its core originality: applying pharmaceutical-level capabilities to the nutraceutical sector. Clients partner with Lonza to secure the trust and stability derived from its strict cGMP manufacturing standards and global regulatory excellence. Rather than

acting as a simple supplier, Lonza drives co-innovation with its customers through R&D hubs like the Innovaform™ Accelerator. The division provides advanced formulations and innovative delivery technologies, such as protecting acid-sensitive ingredients or enhancing bioavailability, which in turn becomes a key competitive advantage for its clients' end products.

5.3. Aenova

Aenova Group is a CDMO – Contract Manufacturer operating in the categories Health & Wellness, Sports & Metabolic, and Beauty & Aesthetic. They are focused on the production of Tablets, Capsules, Powders, as well as Liquids. Their Headquarters are in Germany, however they maintain extensive manufacturing operations in Germany, Switzerland, Italy, Ireland, Romania, and the USA. Aenova is widely recognized as one of Europe's largest and most advanced full-service CDMOs, delivering end-to-end manufacturing solutions across both pharmaceuticals and dietary supplements. The company combines broad technological expertise with a strong regulatory track record, including EU-GMP certification and ISO-compliant production. Its strategic strength lies in its ability to scale complex formulations efficiently while maintaining flexibility for small and mid-sized customers. Aenova's robust infrastructure and technical depth make it a cornerstone of Europe's supplement supply chain. They boast a yearly revenue of approximately €833 million.

Aenova's business model is a B2B CDMO, the company creates value by providing scale manufacturing, regulatory compliance (e.g., EU-GMP, ISO) and formulation support services across 14 sites in Europe and the US. The group serves more than 400 customers, including 7 of the top 10 global pharma companies (Aenova, 2024). This ensures for Aenova a stable and predictable demand as the company's customers rely on Aenova's certified facilities, regulatory compliance and fast product turnaround creating long-term partnerships and reinforcing Aenova's competitive position.

Revenue is captured through contract manufacturing and development service fees across tablets, capsules, powders, and liquids (Aenova, 2024). In 2023, Aenova reported net sales of €833 million, representing a sales growth of 17% YoY. An adjusted EBITDA of €137 million (16.5% EBITDA margin) was reached in 2023, representing a good cost management of the company. Food supplements grew 13% YoY to reach €614m in 2023 accounting for 74% of the total sales of the company.

As an upstream CDMO company, Amazon plays no direct role in Aenova's sales. Even though, their brand customers may sell on Amazon, Aenova doesn't. Customer acquisition and retention in Aenova's model is driven by technical manufacturing of gels, capsules and gummies for example. But it also

driven by regulatory credentials, and large-scale capacity expansions, which together reinforce reliability and stickiness in the Health & Wellness, Sports & Metabolic, and Beauty & Aesthetic categories.

5.4. Sirio Europe (Sirio Pharma)

Sirio Europe (Sirio Pharma) operates as a CDMO within the Health & Wellness, Sports/Metabolic, and Beauty/Aesthetic categories. The division is focused on the production of softgels, gummies, tablets, and probiotics. Its European headquarters is in Germany, established through the acquisition of Ayanda GmbH in 2016 (Sirio Pharma, 2025b). Sirio is positioned as one of the most internationally integrated nutraceutical CDMOs, combining European quality standards with Chinese manufacturing scale. The company is particularly strong in the innovation and production of functional gummies, serving both legacy brands and fast-growing D2C labels. In 2024, Sirio's European regional business reported annual sales of approximately €72.5 million (Sirio Pharma, 2025b).

Sirio's business model is overwhelmingly focused on B2B CDMO solutions, accounting for 99% of its total global revenue in 2024. While the precise dosage form mix for the RMB 604 million in European revenue is not publicly disclosed, it is estimated to be driven by the company's global core competencies. Globally, Sirio's revenue is dominated by Softgels (47.1%) and the high-growth Gummies (23.6%) segment, which together constitute over 70% of its business (Sirio Pharma, 2025b). Given Sirio’s unique technological leadership in gummies, having led the development of the USP standard for CoQ10 gummies this segment is considered the key high-margin growth driver for attracting D2C clients within the European market.

Sirio Europe's customer attraction strategy is based on its core competitive advantage: its Global Hybrid Footprint. European clients can leverage the company's unique combination of European quality, originating from its German R&D hub, and Asian manufacturing speed and scale. This hybrid model provides a powerful proposition for modern D2C brands, which prioritize not only quality but also Speed-to-Market and Cost-Efficiency to navigate fast-moving consumer trends.

5.5. Nitta Gelatin/Wellnex

Nitta Gelatin, operating under the Wellnex brand, is an Ingredient Supplier active in the categories Health & Wellness and Beauty/Aesthetic. They are focused on the production of Collagen Peptides

and Gelatine for beauty-from-within, joint health, and clinical nutrition applications. Their Headquarters are in Japan with manufacturing operations in Japan, the USA, and India. Nitta Gelatin is one of the world's oldest and most reputable producers of collagen and gelatine, combining traditional craftsmanship with advanced bioprocessing technologies. The company is known for its strong commitment to product purity, traceability, and evidence-based innovation. Through its Wellnex portfolio, Nitta provides clinically tested collagen ingredients that are widely used by global supplement and cosmetic brands. The company's long-standing presence in both the Asian and North American markets reinforce its reputation as a trusted supplier within the global collagen industry. (Nitta Gelatin, 2020; Nitta Gelatin, 2022a)

Nitta Gelatin's business model is built around leveraging by-products from the livestock and fisheries industries and transforming them into high-value gelatin, collagen peptide and related specialty materials. The company states that its origins lie in manufacturing industrial gelatin (for example glue) from leather-making by-products in 1918, and it continues today a so-called "up-cycling" business model in which under-utilised raw materials (such as animal skins, bones, fishery by-products) become valuable functional proteins for food, health, beauty and medical applications (Nitta Gelatin, 2020)

Nitta Gelatin operates globally with manufacturing in Japan, India, China, the United States, Canada and Vietnam, enabling its global footprint and supply chain. (Nitta Gelatin, 2022a). The company divides its business into three major segments: Food Solutions, Health Support, and Specialties (sometimes also described as industrial/medical/biomedical). According to the company's profile, the Food Solutions segment includes food gelatin and food materials; the Health Support segment includes gelatin for capsules, collagen peptides for health/beauty and gelatin/collagen for medical uses; the Specialties segment includes adhesives, industrial gelatin and other uses. (Financial Times, n.d.). Nitta Gelatin classifies its sales under three primary categories: Food Solutions, Health Support, and Specialties. Within its FY2024 nine-month results the company reported sales in the "gelatin" product category (formerly part of Food Solutions) of €120.27 million, representing a year-on-year decline of 4.2%. During the same nine-month period ended December 31, 2024, revenue from the "collagen peptide" category increased by 11.8% year-on-year to €28.30 million. (Nitta Gelatin, 2025)

Also, for that period, the "food ingredients" category (gelling agents for desserts, stabilisers for processed meat) generated €13.10 million in net sales, down 2.0% year-on-year. Importantly, the "biomedical" category (gelatin/collagen for medical uses) registered €1.38 million in net sales, marking growth of 32.1% year-on-year. (Nitta Gelatin, 2025). In its full year ended March 31, 2025,

Nitta Gelatin reported consolidated net sales of €213.10 million, gross profit of €54.78 million, and net income of €17.38 million. (Reuters, 2025)

NittaGelatin attracts customers by positioning itself as a high-reliability, customization-driven ingredient partner. The company emphasises its ability to deliver tailor-made gelatin and collagen solutions, a strategy rooted in its dual development approach of creating products both in response to explicit customer needs and through proactive innovation ("need-based" and "seed-based" development). (Nitta Gelatin, 2022a). This capability allows Nitta to serve diverse requirements across food, beauty-from-within, pharmaceutical and biomedical clients. To expand the high-growth collagen peptide category, Nitta invests in advertising and public relations campaigns to raise product awareness and communicate the health benefits of its Wellnex portfolio, directly supporting demand generation. (Nitta Gelatin, 2022a)

In addition, the company uses regional distribution partnerships—such as collaborating closely with distributors in Asia and partnering with Caldic B.V. in Europe, to broaden access to manufacturers across nutrition, personal care and pharmaceutical segments. Nitta further strengthens customer loyalty by leveraging its global application-laboratory network (Japan, U.S., India, China) to co-develop new formulations, enabling faster innovation cycles and increasing switching costs for clients. Combined with its reputation as a top manufacturer, these levers create a consistent pipeline of new customers while reinforcing repeat business. (Caldic, 2024)

5.6. The Role of Amazon in the Food Supplements Industry

This section examines the role of Amazon as a retail platform within the food supplements industry. While the previous section, "Analysis of the Biggest Players" demonstrated that Amazon's direct involvement is limited for large upstream suppliers, understanding its overall impact requires a broader perspective. Therefore, this section also includes players like retailers and final consumers to analyse how Amazon influences product visibility, purchasing behaviour, and competitive dynamics in the industry.

Amazon's influences all three supplement segments: Health & Wellness, Sports & Performance and Beauty & Aesthetics, by acting as both a visibility platform and a primary retail channel. For instance, in 2021 Amazon generated approximately €10.41 billion billion in vitamins, minerals and supplements (VMS) sales (French, 2024). As such, Amazon shapes consumer purchasing behaviour such as brand discovery, review credibility and price comparisons. This phenomenon forces market retailers and smaller D2C brands to adopt platform-centric strategies, this can be reflected in the amount of

supplements brand that we can find on Amazon. The platforms accounts over 10,000 brands in this category and use Amazon to try new products as retail brands can reach customers immediately (Harari in French's work, 2024). The dominance of Amazon has inevitable consequences for upstream supply players. Even if large ingredient suppliers and CDMOs might not sell through Amazon directly, the brands they supply are mostly susceptible to use Amazon influencing demand for supply players.

For all three segments, Amazon helps premium and niche brands reach consumers globally, but the presence of many small brands means product differentiation and packaging become key. Amazon's downstream role is essential, it enables companies to assess what products rise to visibility, for example in the Beauty & Aesthetics segment the keywords that were researched on Amazon for beaty supplements were "Collagen supplements for women", "collagen supplements", "Biotin supplements", "Hair supplements" and "Keratin supplements' (Jungle Scout, 2024). But it also helps assess what packaging, formulations succeed the most, and how quickly product trends diffuse (Harari in French's work, 2024).

From a strategic investment viewpoint, the role of Amazon in the supplements industry is ambivalent. On the one hand, Amazon's dominance ensures that any supplement brand that doesn't use Amazon as a sales channel risks limited reach as the platform is the largest online supplement marketplace (French, 2024). On the other hand, Amazon's marketplace dynamics intensify price transparency and customer acquisition cost (Zhu and Liu, 2018), factors that upstream ingredient suppliers and CDMOs must consider when assessing the financial viability of their downstream customers.

5.7. The Overall Role of Social Media in the Food Supplements Industry

Social media has become a critical driver of awareness, consumer engagement, and ultimately purchase action in the dietary supplement industry. Influencer collaborations, user-generated content (UGC), and algorithm-driven discovery on platforms such as Instagram, TikTok and Facebook enable supplement brands to reach large and engaged audiences in a way that traditional advertising cannot match. For example, research notes that influencer marketing value increased from €1.462 billion in 2016 to an estimated €11.868 billion in 2021, reflecting how brands across consumer health are leveraging social media channels (Klein, 2022).

In the supplement market specifically, a "Study Social Media Performance: Supplements 2024" found that about 90% of the examined supplement providers are active across the six major social media platforms, and roughly 70.5% maintained consistent posting during the study period (Muth, 2024).

In the immune and general health-support segment, social media plays a strong role in education and trend dissemination. According to Quality Matters (2024), 8 in 10 internet users search for health information online and 74% of those users utilise social media as a source. Additionally, influencer content has been shown to shape purchasing decisions in the dietary supplement industry: one survey in Poland found that blogs/vlogs were the main source of product information for 72% of respondents, and 74% said they would look for information after seeing someone else's product they liked. (Roman, 2020)

For the beauty/skin/anti-ageing supplement segment, social media's influence is particularly strong thanks to the visual nature of platforms like Instagram and TikTok, where aspirational imagery, before/after stories and influencer endorsements are highly effective. A content piece on InfluencerMarketingHub (2025) suggests that in the vitamins & supplements industry (which includes beauty-oriented products) the global market is expected to rise from €22.291 billion in 2024 to €37.874 billion by 2032, and brands are increasingly relying on UGC, influencer partnerships and lifestyle positioning on social media. However, the promotional nature of posts can raise regulatory/ethical issues: for example, a study found that Instagram posts advertising skin, hair and nail supplements had only 4% of posts made by verified accounts and only 1% of posts included visible Supplement Facts labels, raising concerns about transparency. (Zamil et al., 2022)

In the sports and performance nutrition segment, social media serves as a platform for community creation, peer review, demonstration of usage (such as gym routines, pre-workout consumption, recovery routines) and peer influence. While specific data for this sub-category are less frequently isolated in published studies, the general findings from Thailand's supplement industry show that structured social media marketing activities significantly correlate with higher customer-commitment metrics in the health-supplement sector. (Nittayoosakulchot & Pankham, 2025). Brands in performance nutrition use social media to display product usage in authentic settings (gym videos, athlete endorsements), leverage micro-influencers in fitness niches, and generate e-word-of-mouth (e-WOM) which influences purchase decisions in this segment. User-generated content (UGC) of workouts, transformations and supplement routines plays especially well here.

5.8. Events in the Food Supplement Industry

Across the main supplement markets in North America, Europe and Asia, the dietary supplement sector convenes at large scale trade fairs that structure business interaction and innovation. In the United States, Natural Products Expo West in Anaheim and SupplySide Global in Las Vegas each gather large communities of health and nutrition professionals and are described as premier trade events for the supplement and health products industries (SupplySide, 2025; WholeFoods Magazine Staff, 2024). Expo West 2024 alone brought more than 65000 industry participants to the Anaheim Convention Center over several days of education, networking and inspirational programming, which illustrates its status as a central meeting point for the natural and organic products community (WholeFoods Magazine Staff, 2024).

In Europe, Vitafoods Europe, now hosted in Barcelona, has evolved into a key reference event for the global nutraceutical and dietary supplement industry, attracting actors from research, formulation, production and distribution (Targeting Gut Disease S R L, 2024). Italy further consolidates its central role in the European supplement market through in Vitality in Milan, described as a leading business to business event for the dietary supplement sector (Encanto Nutraceutica, n.d.). These European trade shows act as regional hubs for innovation and commerce where firms present novel ingredients, delivery formats and functional foods to an international buyer base (Encanto Nutraceutica, n.d.; Targeting Gut Disease S R L, 2024).

In the Asia Pacific region, Healthplex Expo Natural and Nutraceutical Products China in Shanghai functions as a core platform for brands and manufacturers that seek to expand markets, track industry trends, develop distribution channels and keep pace with regulatory change (PharmaLinea, 2025). In India, Vitafoods India in Mumbai has quickly become a central nutraceutical meeting place, with the third edition in 2025 featuring 159 exhibitors and an expected footfall of more than 8000 visitors, positioning itself as a premier networking and knowledge sharing platform for the nutraceutical ecosystem (Informa Markets in India, 2025).

These gatherings are more than routine trade fairs. They are mechanisms that support industry growth and the diffusion of innovation. Global expos provide launch platforms for new products and ingredients, which allows novel supplement trends to move quickly across markets as firms present them in front of concentrated groups of buyers, suppliers and media representatives (SupplySide, 2025; WholeFoods Magazine Staff, 2024). Within this dense networking environment, executives, formulators and distributors meet face to face to form relationships and negotiate partnerships and distribution agreements that underpin international expansion (PharmaLinea, 2025; Vitafoods Global,

2025). The scale of participation at these expos therefore serves as a visible signal of where investment and strategic interest are concentrating within the global dietary supplement industry.

Although these expos mainly serve business audiences rather than end consumers, they still influence consumer awareness and demand. Trade press and digital media coverage around events often highlight popular categories and ingredients, which informs consumers and retailers and gives additional legitimacy to emerging products (Informa Markets in India, 2025; Targeting Gut Disease S R L, 2024; WholeFoods Magazine Staff, 2024). Overall, prominent supplement expos shape the global trajectory of the dietary supplement sector, by facilitating intensive international networking, brand exposure and competition.

5.9. Role of Renowned Chefs in the Food Supplements Industry

Globally recognised chefs have become increasingly influential in the nutraceutical and dietary supplement space, effectively connecting high end gastronomy with health science. Their endorsements and co-branded collaborations provide an additional layer of culinary credibility to supplement brands, strengthening perceived quality, authenticity, and safety in the eyes of consumers. Research on celebrity chefs and destination marketing shows that chefs transfer their expertise, trustworthiness, and aesthetic authority to the products they endorse, improving brand image and willingness to try new offerings (Demirkol & Cifci, 2020). Industry analyses similarly emphasise that celebrity chef partnerships allow food and nutrition brands to leverage the chef's reputation and highly engaged audience, which elevates the perceived premium nature and desirability of endorsed products (DML Talent, n.d.).

One clear manifestation of this logic is the use of chefs as flavour specialists for supplements. Vitafusion, for example, appointed viral chef Brad Leone as its Chief Flavor Officer to help design and promote chef inspired gummy multivitamins, marketed as combining strong taste with complete nutritional value (New York Post, 2025). This type of collaboration uses chefs' expertise in taste, texture, and ingredient balancing to address one of the structural weaknesses of many supplements, namely poor palatability. By improving the sensory profile of vitamins and nutraceutical products, chefs make adherence more likely and transform supplementation from a chore into an experience aligned with contemporary food culture (New York Post, 2025). At the same time, the communication campaign around the partnership capitalises on Leone's existing community and media presence,

creating conversational momentum in social channels that reinforces consumer trust and social proof (DML Talent, n.d.).

Renowned chefs also contribute directly to product innovation by applying fine dining level creativity to nutritionally enhanced products. A striking illustration is the collaboration between Michelin starred chef Miller Prada and the luxury protein brand Hermosa, which resulted in a visually elaborate and high-priced protein shake served in premium fitness studios (Gecsoyler, 2025). This approach exemplifies a broader trend in which chefs experiment with protein powders, superfoods, and botanical extracts to create recipes and product formats that are both nutritionally dense and sensorially attractive, thereby narrowing the gap between gastronomy and nutraceuticals (Gecsoyler, 2025).

Beyond product formulation, celebrity chefs act as educators and cultural intermediaries who translate nutritional science into everyday practice. Through cookbooks, television, online content, and live events, they teach consumers how to use nutrient rich ingredients and, increasingly, how to integrate functional foods and supplements into daily routines. Chef David Bouley's The Chef and the Doctor series, for example, stages dinners where medical experts present evidence on the health impacts of specific ingredients, followed by multi course menus built around those components (Party Digest, n.d.). The format explicitly promotes a food as medicine narrative and positions functional ingredients, including those commonly found in supplements, as tools for disease prevention and wellness optimisation. Such initiatives strengthen public understanding of nutrition and reinforce the legitimacy of science backed ingredients in both food and supplement form (Party Digest, n.d.; Jamie Oliver Group, 2020).

Celebrity chefs are widely recognised trend setters in food culture, and their influence clearly extends to ingredients that straddle the boundary between food and supplementation. The rapid rise of kale as a mass market "super vegetable" illustrates how chef and celebrity endorsements can trigger measurable shifts in retail demand (Smithers, 2014). This pattern is consistent with research showing that celebrity experts shape consumer attitudes and behaviour by lending credibility, aspirational identity, and attention to specific products or ingredients (Demirkol & Cifci, 2020). Popular figures such as Jamie Oliver and Joe Wicks have similarly used books, campaigns, and digital content to reframe healthy eating as accessible and desirable, linking cooking skills, nutrient intake, and mental wellbeing and thereby raising interest in the vitamins and functional components behind their recipes (Jamie Oliver Group, 2020; West, 2022).

Overall, globally recognised chefs have emerged as strategic actors in the evolution of the dietary supplement and broader nutraceutical market. Through endorsements and co branded collaborations

they enhance perceived quality and trust in supplement products and reinforce premium positioning (Demirkol & Cifci, 2020; DML Talent, n.d.). Through education and agenda setting in media and live events they diffuse food as medicine narratives, shape ingredient trends, and encourage consumers to experiment with functional foods and supplements as part of everyday health routines (Jamie Oliver Group, 2020; Party Digest, n.d.; Smithers, 2014; West, 2022). In combination, these mechanisms mean that chefs are no longer only creators of restaurant dishes, they have become influential intermediaries who help set global wellness and nutrition trends and indirectly accelerate the adoption of dietary supplements.

6. Scenario Analysis

Building on the preceding demand, taxonomy, and supply side analysis, this scenario section translates our research into concrete investment options for the fund. We identify four ranked recommendations that differ by business model (contract versus integrated manufacturing), supplement subcategory (Health and Wellness; Sports, Performance and Metabolic; Beauty and Aesthetic) and geography (Europe; Asia-Pacific, America). Each scenario is evaluated against market trends, market maturity, structural growth rates, and the practical advantages of cultural and operational proximity. The core question is how to best capture the structural expansion of the supplements market, which is growing at roughly 7-9% annually globally (Statista, Total dietary supplements market size worldwide from 2018 to 2028, 2025a; Grand View Research, 2025a) while managing risk and execution complexity for a European private equity investor.

6.1. Scenario 1 - Contract manufacturing as the primary investment avenue

Across all business models, contract manufacturing emerges as the most attractive locus for value creation. Global dietary supplements represent a market of about €165.68 billion with an expected CAGR of 8.9%, whereas the contract manufacturing segment alone already accounts for €51.28 billion and is projected to grow at 12.6% annually through 2030 (Grand View Research, 2025a, 2025b). In other words, the upstream CDMO pool is expanding significantly faster than the overall category. This growth is structurally underpinned by high regulatory and capital barriers to in house production, which smaller and mid sized brands increasingly circumvent by outsourcing to specialist CDMOs (Eubico, 2025; Millmax, n.d.).

Additionally, the economics of leading CDMOs show that this model can combine scale with attractive profitability. Aenova, Europe's largest provider of solid and semi solid supplements, reported 2023 net sales of €833 million, up 17% year on year, with adjusted EBITDA of €137 million and a 16.5% margin; food supplements contributed €614 million, or 74% of total sales (Aenova, 2025). This indicates that CDMOs can capture a sizable and growing share of the category's value chain while remaining asset backed, cash generative and defensible due to GMP and ISO certified capabilities. On this basis, our primary scenario directs new PE capital first towards contract manufacturing, where the fund can gain leverage on the whole industry's expansion rather than betting on individual brands.

6.1.1. Scenario 1.a - European Health and Wellness CDMO exposure

Our highest priority recommendation is an equity position in a European CDMO with a strong Health and Wellness focus. Europe's dietary supplements market was worth €23.8 billion in 2019 and is expected to reach €44 billion by 2026, implying a 9.5% CAGR, which is higher than both North America and Asia-Pacific (Djaoudene et al., 2023). Italy alone exceeds €1.6 billion in supplement sales, supported by a pharmacy led distribution model and high consumer trust in multivitamins, digestive health and herbal products (Farmaimpresa, 2024). This regional profile combines robust growth with relatively high regulatory standards and strong healthcare channels, which is advantageous for an upstream manufacturing play.

Within this regional context, Health and Wellness supplements are the largest and most resilient subcategory. Globally this segment is valued at about €80 billion in 2024 and set to reach €116 billion by 2030 at a 6.4% CAGR (Statista, 2025b). Around two thirds of supplement users take a vitamin or mineral product, confirming that Health and Wellness sits at the behavioural core of the category (Djaoudene et al., 2023).

A European CDMO positioned in this Health and Wellness pool benefits from three reinforcing effects: structurally higher regional growth, a very large and diversified demand base in vitamins, minerals, herbal and plant-based products, and the technical/regulatory moat created by EU GMP, ISO 9001 and related quality systems (Aenova, 2025). For a European fund, cultural and geographic proximity additionally reduces execution risk in governance, operational oversight and eventual exit.

6.1.2. Scenario 1.b - Asia-Pacific Beauty and Aesthetic CDMO exposure

The second ranked contract manufacturing option targets the intersection of Asia-Pacific growth and the Beauty and Aesthetic supplements segment. The Asia-Pacific dietary supplements market has expanded from approximately €29.1 billion in 2018 to a projected €48.4 billion by 2026, corresponding

to a CAGR of about 6.5% (Statista, Value of the dietary supplements market worldwide by region, 2018–2026, 2018). This trajectory is driven by rising disposable income, an expanding middle class, and strong traditions of functional nutrition in countries such as Japan and China (Djaoudene et al., 2023).

Within this regional base, beauty focused ingestible supplements represent the fastest growing subcategory in our taxonomy. Global market value for beauty supplements exceeded €6 billion in 2024 and is forecast to grow at over 8% CAGR through 2030, nearly doubling from around €3.0 billion in 2016 (Statista, 2025d). Signals from Southeast Asia show that beauty objectives have become mainstream. In 2024, beauty supplements accounted for 26.8% of the health supplements category in the region (Statista, Market share of health supplements in Southeast Asia in 2024, by category, 2025i).

An Asia-Pacific CDMO specialising in collagen, hyaluronic acid and related beauty ingredients can therefore capture a high growth niche that is rapidly normalising into everyday routines. For the fund, such an exposure offers higher CAGR than a comparable Americas strategy, at the cost of greater distance and regulatory complexity. This trade off justifies its position as the second, rather than primary, contract manufacturing scenario.

6.2. Scenario 2 - Integrated manufacturing as a secondary but attractive path

Integrated manufacturers combine in house production with owned consumer brands and direct distribution, often leveraging “farm to capsule” narratives of quality and traceability (Amway, n.d.; Nature’s Sunshine, 2024). While these businesses typically achieve higher top line revenues than CDMOs, they are more exposed to brand, channel, and consumer preference risk. In addition, they do not benefit from the natural diversification that CDMOs enjoy across hundreds of B2B clients. For this reason we view integrated manufacturing as a secondary, though still compelling, route for the fund, particularly where brand equity and regional dynamics are strongly aligned with underlying structural growth.

6.2.1. Scenario 2.a - European Health and Wellness integrated brands

Our first integrated scenario again prioritises Europe and the Health and Wellness segment, but this time on the brand owner side. The same macro conditions that make Europe attractive for CDMOs also support the economics of integrated players. Health and Wellness remains the largest and most widely consumed subcategory globally, with €80 billion in value in 2024 and a forecast €116 billion by 2030 (Statista, Global health and wellness supplements market size from 2019 to 2030, 2025b).

European consumers frequently use multivitamins, digestive health and herbal formulations for prevention and everyday wellbeing (Djaoudene et al., 2023).

An integrated European manufacturer that owns both the production assets and the brand can therefore capture margin at both manufacturing and retail levels, while benefiting from pharmacy led recommendation and rising per capita spending on health and fitness supplements (Statista, Average annual per capita spending on health and fitness supplements in selected European countries, 2025f). With an appropriate mix of omnichannel distribution and innovation in formats such as gummies and ready to drink products, such a platform could represent a robust cash generator with multiple levers for operational value creation. However, because the upside is more tightly linked to brand execution than in a CDMO model, we place this scenario below contract manufacturing in the ranking.

6.2.2. Scenario 2.b - Asia-Pacific Beauty and Aesthetic integrated brands

The final scenario focuses on integrated beauty supplement brands in Asia-Pacific. As noted above, the region's supplements market is growing at approximately 6.5% annually, supported by increasing incomes and strong traditions of functional nutrition (Statista, Value of the dietary supplements market worldwide by region, 2018–2026, 2018; Djaoudene et al., 2023). At the same time, global beauty supplements have nearly doubled in value between 2016 and 2024 and are projected to expand at over 8% CAGR through 2030 (Statista, Estimated market value of beauty supplements worldwide in 2016 and 2024, 2025d; Statista, Beauty supplement market size worldwide from 2023 to 2030, 2025). In Southeast Asia, beauty already accounts for more than a quarter of the health supplements spend (Statista, Market share of health supplements in Southeast Asia in 2024, by category, 2025i).

Integrated beauty brands in this region benefit from three mutually reinforcing drivers: the convergence of beauty and wellness culture, subscription like usage patterns for collagen and skin formulas, and highly digital go to market models built on social media and influencer commerce (Euromonitor International, 2024; Business of Fashion & McKinsey, 2025). A Chinese or Southeast Asian integrated beauty supplement platform can translate these structural drivers into significant top line and brand equity, similar to how By Health has scaled within the broader supplements category. For a European fund, such an investment would add geographic and category diversification with a strong growth narrative, while also introducing higher execution complexity in regulation, cross border governance and consumer insight.

Overall, we therefore position integrated Asia-Pacific beauty as the fourth ranked scenario: attractive from a growth and story perspective, but less risk efficient than the preceding three options. The fund's

core recommendation remains to prioritise European Health and Wellness CDMOs, followed by targeted exposure to Asia-Pacific beauty CDMOs, and only then to consider integrated platforms that offer sufficiently strong brand, governance and exit prospects to justify their higher idiosyncratic risk.

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Appendix 1.

Table 4. Overview of General Biggest Market Players with Certifications and Sources

Company	Segment (Ingredient / CDMO / Capsules & Delivery Tech)	Estimated Revenue (FY 2023–24)	Category Fit (H&W; Sports/Metabolic; Beauty/Aesthetic)	Core Formats/ Tech	Geography - HQ	Geography - Manufacturing Footprint	Certs / Notes	Source
Aenova	CDMO–contract manufacturer	≈ €833 million	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Tablets, Capsules, Powders, Liquids	Germany	Germany, Switzerland, Italy, Ireland, Romania, USA	EU-GMP, ISO 9001; full-service manufacturing	(Aenova Group, 2025)
Sirio Europe (Sirio Pharma)	CDMO–contract manufacturer	≈ **€436.9 million** (FY 2023)	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Softgels, Gummies, Tablets, Probiotics	Germany	China, USA, Germany	GMP, ISO 22000; global nutraceutical CDMO	(Sirio Europe, 2025)
Lonza Capsules & Health Ingredients	Capsules & delivery tech	€1.03 billion	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Capsules, HPMC Capsules, Probiotics	Switzerland	Switzerland, USA, China, Japan, India	cGMP, NSF, ISO; advanced capsule technologies	(Lonza Group, 2023)
DSM-Firmenich	Ingredient supplier	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Premixes, Vitamins, Nutraceuticals	Switzerland / Netherlands	Global (EU, US, APAC)	ISO, FSSC 22000; vertically integrated nutrition solutions	(DSM-Firmenich, 2025)
Rousselot (Peptan)	Ingredient supplier (collagen)	≈ €1.46 billion (2023, Darling Ingredients Food Ingredients segment)	Health & Wellness, Beauty/Aesthetic	Collagen Peptides	France	France, Belgium, Brazil, China, USA	Upcycled Certified®, ISO, FSSC 22000	(Rousselot, 2025) (Darling Ingriedients Inc.,2023)
Millmax	CDMO–contract manufacturer	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Capsules, Tablets, Powders, Gummies	Poland	Poland	ISO 22000, GMP; multi-category supplement CDMO	(Millmax, 2025)
Eubioco	CDMO–contract manufacturer	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Tablets, Capsules, Powders, Sachets	Poland	Poland	GMP, ISO 9001; supplement contract manufacturing	(Eubioco, 2025)
BASF Nutrition & Health	Ingredient supplier	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Vitamins, Carotenoids, Plant Sterols	Germany	Germany, Global network	ISO, GMP; vitamin and carotenoid ingredients	(BASF Nutrition & Health, 2025)
Vitaquest	CDMO–contract manufacturer	≈ €215–258 million (estimate)	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Capsules, Tablets, Powders, Gummies	USA	USA (NJ facilities)	GMP, NSF, ISO; turnkey nutraceutical CDMO	(Vitaquest, 2025)
NutraScience Labs	CDMO–contract manufacturer	≈ €86–129 million (estimate)	Health & Wellness, Sports/Metabolic	Capsules, Powders, Gummies, Tablets	USA	USA	GMP, FDA-registered facility	(NutraScience Labs, n.d.)
Kaneka Nutrients	Ingredient supplier	Non-public	Health & Wellness, Sports/Metabolic	CoQ10, Ubiquinol	Japan / USA	USA (Texas), Japan	GMP, ISO 9001; branded CoQ10/Ubiquinol ingredients	(Kaneka Nutrients, 2025)
Glanbia Nutritionals	Ingredient supplier	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Proteins, Bioactives, Premixes	Ireland / USA	Ireland, USA, China	ISO, FSSC 22000; major global ingredient supplier	(Glanbia Nutritionals, 2025)
TSI Group	CDMO–contract manufacturer	≈ €172–215 million (2023 estimate)	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Capsules, Powders, Tablets, Custom Formulations	China / Australia	China, Australia, Thailand, EU, USA	GMP, ISO; multi-region nutraceutical manufacturing	(TSI Group, 2025)
Novonesis (Novozymes + Chr. Hansen)	Ingredient supplier	Non-public	Health & Wellness, Beauty/Aesthetic	Probiotics, HMOs, Enzymes	Denmark	Denmark, USA, Global	GMP, ISO, FSSC 22000; biotechnology-based ingredients	(Novonesis, 2025)
Kerry ProActive Health	Ingredient supplier	Non-public	Health & Wellness, Sports/Metabolic, Beauty/Aesthetic	Probiotics, Postbiotics, Bioactives	Ireland	Global	ISO, FSSC 22000; gut and immune health ingredients	(Kerry Group, 2025)
Nitta Gelatin / Wellnex	Ingredient supplier (collagen & gelatin)	€308 million	Health & Wellness, Beauty/Aesthetic	Collagen Peptides, Gelatin	Japan	Japan, USA, India	ISO, FSSC 22000; collagen and gelatin manufacturer	(Nitta Gelatin Inc., 2023)

Table 5. Overview of Biggest Market Players – Retail Focus with sources

Company	Supplement Revenue (Latest)	Manufac-turing activity	Manufacturing Locations	Retail Focus	Additional Information	Category Fit	Headquarters	Sources
Nestlé Health Science	€5.59B (2024)	Yes	USA, Canada, France, Germany, Brazil, China, Switzerland	Global retail brands, pharmacies, e-commerce, grocery	Owns Garden of Life, Vital Proteins, Pure Encapsulations	Health & Wellness, Beauty/Aesthetic, Sports/Metabolic	Vevey, Switzerland	(Nestlé S.A., 2024)
Herbalife Nutrition	€4.27B (2024)	Yes	USA, Mexico, Brazil, China, India	Direct-to-consumer retail, global distributors	Global sports & active nutrition brand	Health & Wellness, Sports/Metabolic	Los Angeles, USA	(Herbalife Ltd., 2024)
Amway (Nutrilite)	€2.92–3.18B (2024)	Yes	USA (California, Washington), Brazil, Mexico, India, China	D2C/MLM retail distribution	Vertically integrated; operates own farms	Health & Wellness, Sports/Metabolic	Ada, Michigan, USA	(Amway, 2024)
Haleon (Centrum, Caltrate)	€2.67B (2024, VMS only)	Yes	USA, UK, Ireland, China, India	Pharmacies, mass retail, supermarkets	Large global consumer-health portfolio	Health & Wellness	Weybridge, United Kingdom	(Haleon plc, 2024)
Bayer VMS	€1.46B (2024, VMS only)	Partially	Germany, USA, Spain, South America	Pharmacies, grocery, mass retail	Brands include One A Day, Berocca, Elevit	Health & Wellness	Leverkusen, Germany	(Bayer AG, 2024)
Pharmavite (Nature Made)	€1.29B (2024)	Yes / Partly	USA (California, Ohio)	Pharmacies, mass retail, e-commerce	Top US vitamin brand owned by Otsuka	Health & Wellness	West Hills, California, USA	
Church & Dwight (Vitafusion)	€946M (2024, VMS)	Yes	USA	Mass retail, pharmacies, grocery	Leader in gummy supplements	Health & Wellness	Ewing, New Jersey, USA	(Church & Dwight Co., Inc., 2024)
Abbott Nutrition (Supplements)	€1.20B (2024 est.)	Yes	USA, Ireland, Netherlands, Spain, Singapore	Pharmacies, grocery, clinical retail	Includes Ensure and Pedialyte	Health & Wellness	Abbott Park, Illinois, USA	(Abbott Laboratories, 2024)
NOW Foods	€645M (2024)	Yes	USA (Illinois, Nevada)	Natural retailers, mass retail, e-commerce	Large clean-label supplement manufacturer	Health & Wellness, Sports/Metabolic	Bloomingdale, Illinois, USA	(NOW Health Group, 2024)
Swisse (H&H Group)	€499M (2024)	No	Manufactured via CDMOs in China, Australia, EU	Premium retail, pharmacies, e-commerce	Beauty and wellness positioning	Beauty/Aesthetic, Health & Wellness	Hong Kong / Melbourne, Australia	(H&H Group, 2024)

Appendix 2.

Academic Integrity Statement on AI Usage

In the preparation of this report, AI-assisted tools have been utilized to enhance linguistic clarity, grammar, spelling, and overall coherence. These tools have been employed solely as a means of refining expression and ensuring professional readability, without altering the substantive content or analytical rigor of the work.

We acknowledge that AI can be a valuable aid in improving written communication; however, we emphasize that all intellectual contributions, research, analysis, and arguments presented in this report are our own. The use of AI does not replace critical thinking, academic diligence, or personal accountability.

As the authors, we take full responsibility for the accuracy, validity, and originality of this work. The report adheres to the highest standards of academic integrity, and any AI-generated refinements have been carefully reviewed to ensure they align with the intended meaning and ethical principles of scholarly writing.